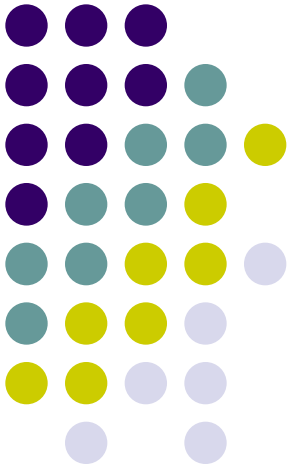
OSBIS Project Week2

Presented by Victor Huo

Oct 16, 2022

Team 1



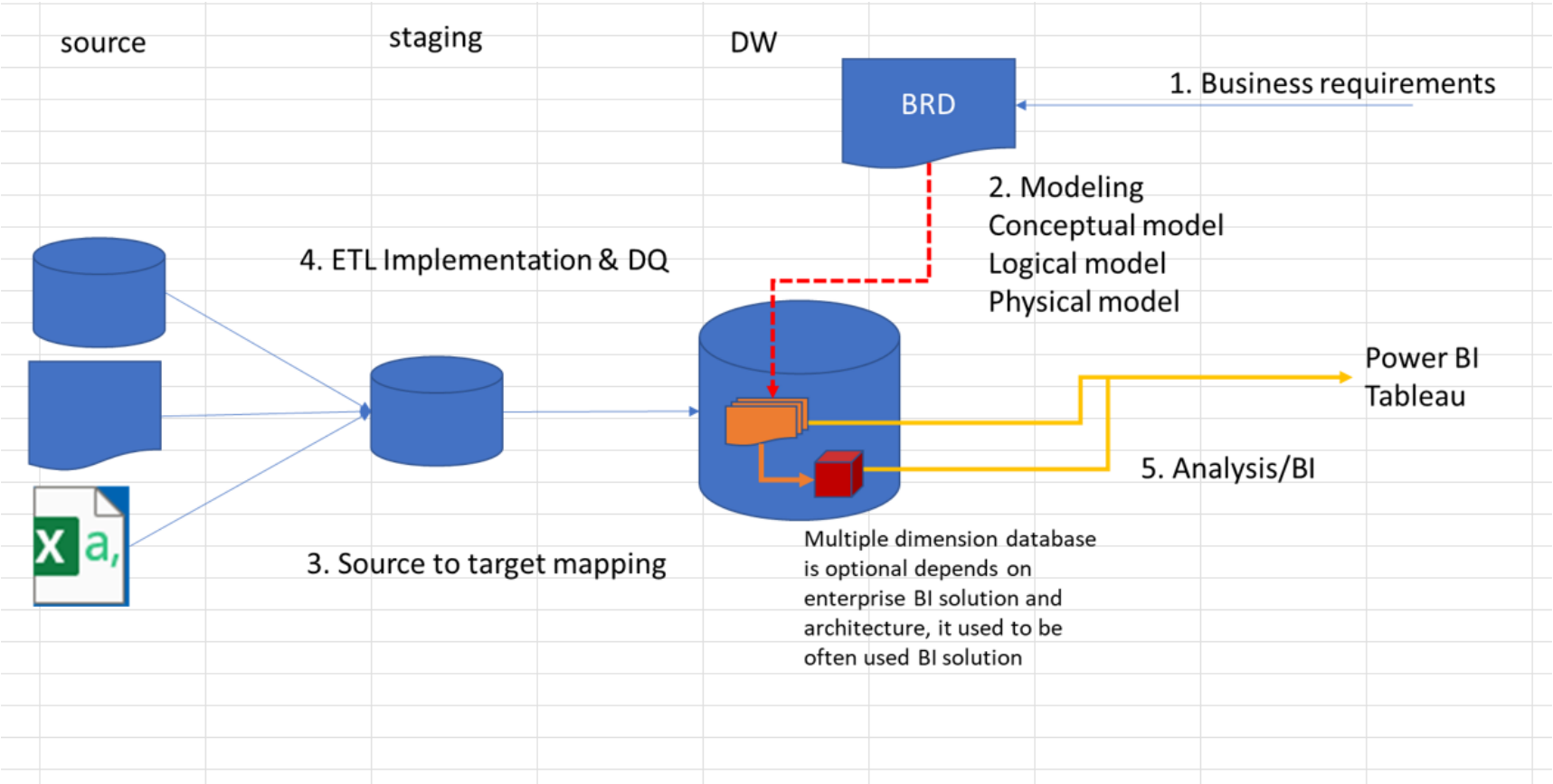
Agenda

- Project Background
- Data Warehousing Project Process
- EDW Data Mart Modeling
- Source to Target Mapping
- ETL with SSIS – Load DimCustomer

Project Background

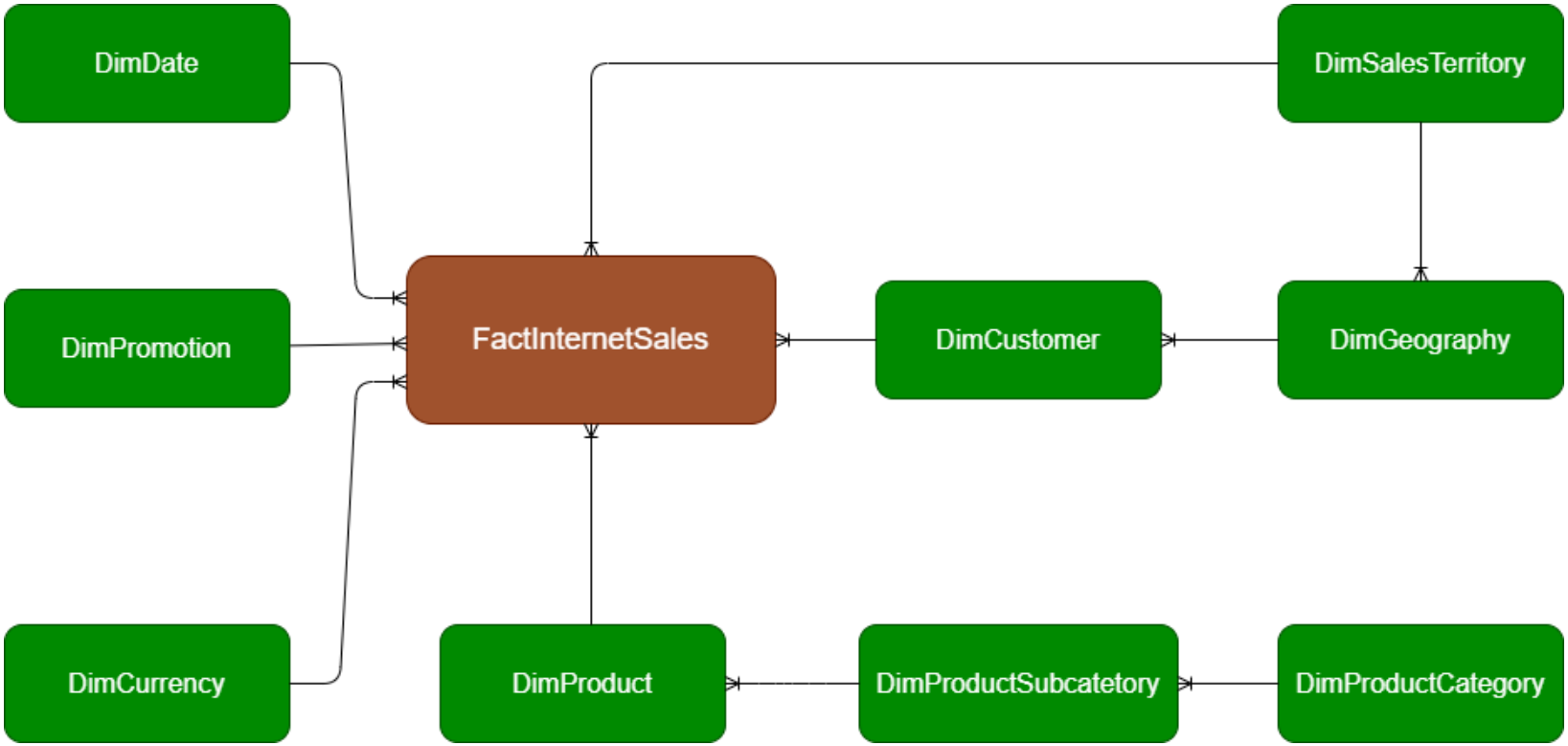
- Adventure Works Cycles is a fictitious multinational bicycle manufacturer.
- The company has two main revenue streams: Internet Sales and Reseller Sales. This data analysis project will be focused on internet sales.
- To perform enterprise-wide online sales analysis, management decides to build a data warehouse (EDW) to integrate all sales related data into a central repository.
- Data mart is a smaller version of EDW which deals with a single subject. We need to build a data mart for internet sales.

Data Warehousing Project Process

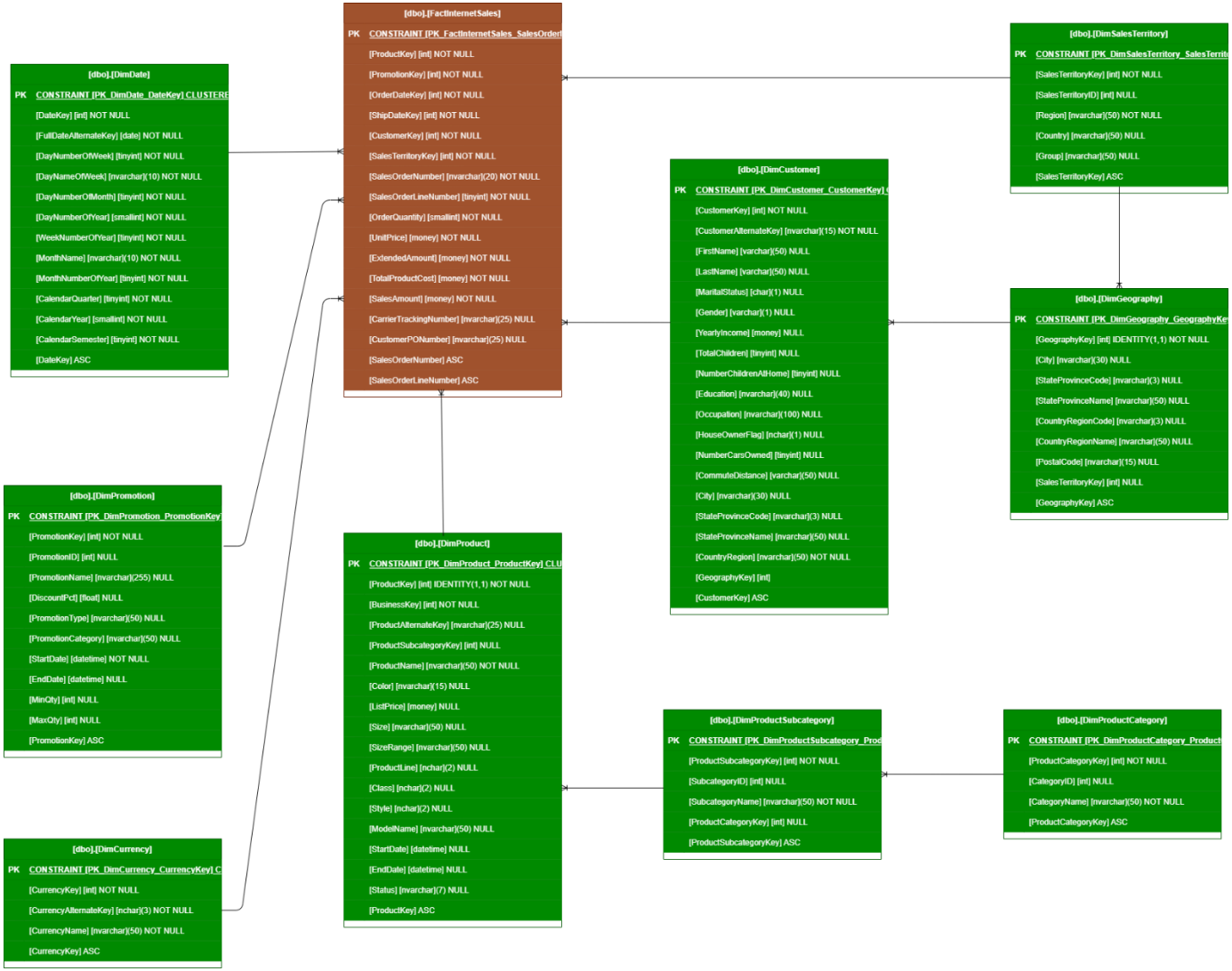


Data Mart Modeling - Conceptual Model

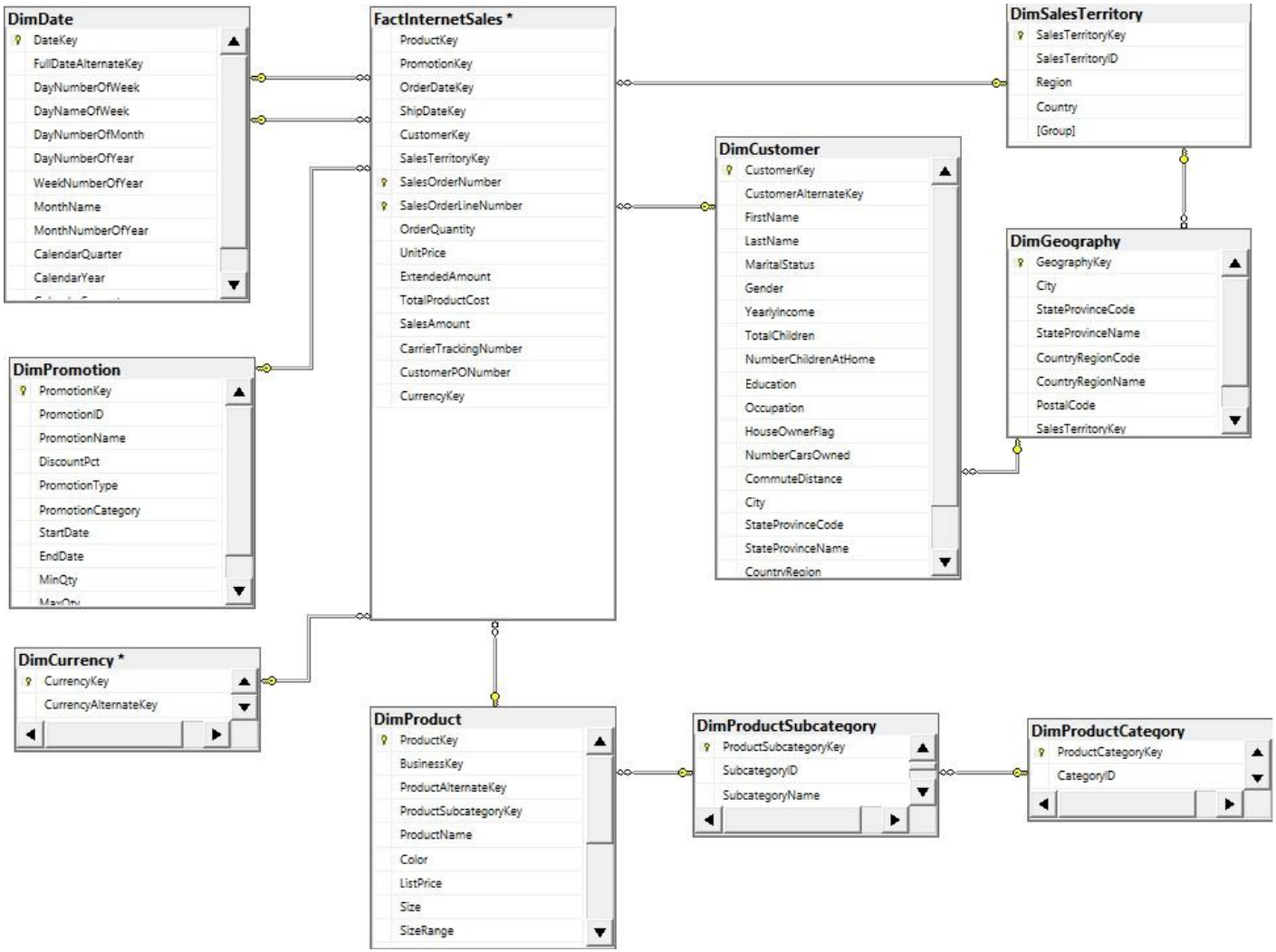
- Snowflake Schema



Data Mart Modeling - Logical Model



Data Mart Modeling - Physical Model



Source to Target Mapping – Create views

- Source database - AdventureWorks2019
- Target database - AdventureWorksDW
- Create the following views in the source database to facilitate mapping and ETL process
  - dbo.etl_vCustomerDemographics
  - dbo.etl_DimCustomer
  - dbo.etl_DimProduct
  - dbo.etl_Geography
- Create stage.DimCustomer in the target database

Source to Target Mapping –
Dimension Table DimCustomer

Source (AdventureWorks2019)	Target (AdventureWorksDW)
Person.Person Person.Address Person.BusinessEntityAdress Person.EmailAdress Person.PersonPhone Person.StateProvince Sales.Customer Person.CountryRegion dbo.etl_vCustomerDemographics	DimCustomer
Source (AdventureWorksDW)	
DimGepgraphy	

Source to Target Mapping –
Dimension Table DimCustomer

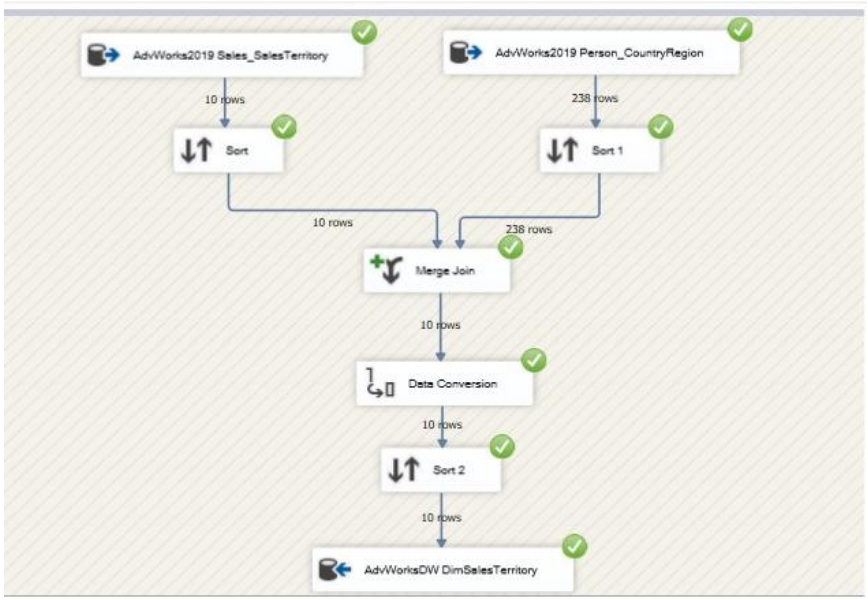
Table Name	DimCustomer														
Table Type	Dimension														
Display Name	Customer Dimension														
			Target									Source				ETL
Database	Schema	Table Name	Column Name	Data Type	Size	Key?	FK To	NULL?	Default	Database	Schema	Table/View Name	Column Name	Data Type	Size	ETL Rules
AdventureWorksDW	dbo	DimCustomer	CustomerKey	int		PK		NO		AdventureWorks2019	Sales	Customer	CustomerID	int		Get from source
AdventureWorksDW	dbo	DimCustomer	CustomerAlternateKey	nvarchar	15			NO		AdventureWorks2019	Sales	Customer	AccountNumber	varchar	10	Data conversion
AdventureWorksDW	dbo	DimCustomer	GeographyKey	int		FK	DimGeogr	YES		AdventureWorksDW	dbo	DimGeography	GeographyKey			Get from DimGeography
AdventureWorksDW	dbo	DimCustomer	FirstName	varchar	50			YES		AdventureWorks2019	Person	Person	FirstName	nvarchar	50	Data conversion
AdventureWorksDW	dbo	DimCustomer	LastName	varchar	50			YES		AdventureWorks2019	Person	Person	LastName	nvarchar	50	Data conversion
AdventureWorksDW	dbo	DimCustomer	MaritalStatus	char	1			YES		AdventureWorks2019	Sales	vCustomerDemographic	MaritalStatus	nvarchar	1	Data conversion
AdventureWorksDW	dbo	DimCustomer	Gender	varchar	1			YES		AdventureWorks2019	Sales	vCustomerDemographic	Gender	nvarchar	1	Data conversion
AdventureWorksDW	dbo	DimCustomer	YearlyIncome	money				YES		AdventureWorks2019	Sales	vCustomerDemographic	YearlyIncome	nvarchar	30	Dervied from ranges
AdventureWorksDW	dbo	DimCustomer	TotalChildren	tinyint				YES		AdventureWorks2019	Sales	vCustomerDemographic	TotalChildren	int		Direct move
AdventureWorksDW	dbo	DimCustomer	NumberChildrenAtHome	tinyint				YES		AdventureWorks2019	Sales	vCustomerDemographic	NumberChildrenAtHom	int		Direct move
AdventureWorksDW	dbo	DimCustomer	Education	nvarchar	40			YES		AdventureWorks2019	Sales	vCustomerDemographic	Education	nvarchar	30	Direct move
AdventureWorksDW	dbo	DimCustomer	Occupation	nvarchar	100			YES		AdventureWorks2019	Sales	vCustomerDemographic	Occupation	nvarchar	30	Direct move
AdventureWorksDW	dbo	DimCustomer	HouseOwnerFlag	nchar	1			YES		AdventureWorks2019	Sales	vCustomerDemographic	HouseOwnerFlag	bit		Direct move
AdventureWorksDW	dbo	DimCustomer	NumberCarsOwned	tinyint				YES		AdventureWorks2019	Sales	vCustomerDemographic	NumberCarsOwned	int		Direct move
AdventureWorksDW	dbo	DimCustomer	CommuteDistance	varchar	50			YES		Derived						
AdventureWorksDW	dbo	DimCustomer	City	nvarchar	30			YES		AdventureWorks2019	Person	Address	City	nvarchar	30	Direct move
AdventureWorksDW	dbo	DimCustomer	StateProvinceCode	nvarchar	3			YES		AdventureWorks2019	Person	StateProvince	StateProvinceCode	nvarchar	3	Direct move
AdventureWorksDW	dbo	DimCustomer	StateProvinceName	nvarchar	50			YES		AdventureWorks2019	Person	StateProvince	Name	nvarchar	50	Direct move
AdventureWorksDW	dbo	DimCustomer	CountryRegion	nvarchar	50			NO		AdventureWorks2019	Person	CountryRegion	Name	nvarchar	50	Direct move

Source to Target Mapping –
DimCustomer's Referenced Tables

Source (AdventureWorks2019)	**Target (AdventureWorksDW)**
Sales.SalesTerritory	DimSalesTerritory
Person.CountryRegion	
Source (AdventureWorks2019)	DimGeography
Person.Address Person.StateProvince Person.CountryRegion	
Source (AdventureWorksDW)	
DimSalesTerritory	

ETL with SSIS – Load Referenced Tables DimSalesTerritory and DimGeography

- Create a SSIS project AdvWorksDW ETL
- Create SSIS packages for relevant dimension tables
- Execute SSIS packages to load relevant dimension tables



ETL with SSIS – Load DimCustomer

- 72 duplicate records in dbo.etl_DimCustomer to be removed

```sql
USE AdventureWorks2019
GO

SELECT COUNT(CustomerID) AS TotalRows
FROM [dbo].[etl_DimCustomer]

SELECT COUNT(DISTINCT CustomerID) AS DistinctRows
FROM [dbo].[etl_DimCustomer]
```

100 %

Results | Messages

	TotalRows
1	18556

	DistinctRows
1	18484

Query executed successfully.

```sql
WITH cte AS (
    SELECT
        CustomerID,
        ROW_NUMBER() OVER (
            PARTITION BY
                CustomerID
            ORDER BY
                CustomerID
        ) row_num
    FROM
        dbo.etl_DimCustomer
)
SELECT * FROM cte
WHERE row_num > 1;
```

100 %

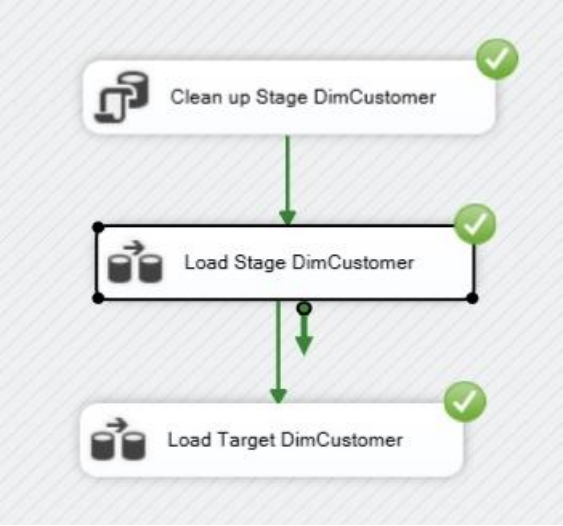
Results | Messages

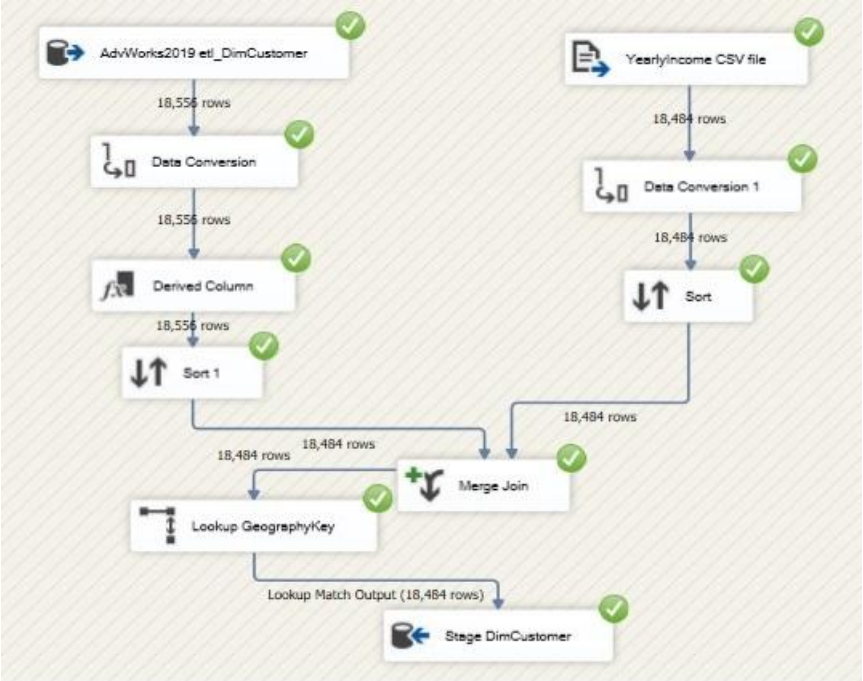
	CustomerID	row_num
1	11279	2
2	11279	3
3	11279	4
4	11532	2
5	11532	3
6	11532	4
7	13145	2
8	13145	3
9	13145	4
10	13311	2
11	13311	3
12	13311	4

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (66) | AdventureWorks2019 | 00:00:01 | 72 rows

ETL with SSIS – Load DimCustomer

- Removing duplicated records via Sort
- Getting the YearlyIncome and CommuteDistance by Merge Join with YearlyIncome CSV file
- Lookup GeographyKey from DimGeography





OSBIS Project Week3

Presented by Victor Huo

Oct 23, 2022

Team 1

Agenda

- Source to Mapping
- Load DimProduct
- Load DimDate
- Load DimCurrency
- Load DimPromotion
- Load FactInternetSales
- Questions

Source to Target Mapping

- For DimProduct

```sql
USE AdventureWorks2019
GO
CREATE VIEW dbo.etl_DimProduct AS
SELECT        ppm.Name AS ModelName, pp.DiscontinuedDate, pp.SellEndDate, pp.SellStartDate, pp.ProductModelID,
                    pp.ProductSubcategoryID, pp.Style, pp.Class, pp.ProductLine, pp.DaysToManufacture, pp.Weight,
                    pp.WeightUnitMeasureCode, pp.SizeUnitMeasureCode, pp.Size,
              CASE WHEN pp.Size='38' OR pp.Size='39' OR pp.Size='40'  THEN '38-40 CM'
                  WHEN pp.Size='42' OR pp.Size='43' OR pp.Size='44' OR pp.Size='45' OR pp.Size='46' THEN '42-46 CM'
                  WHEN pp.Size='48' OR pp.Size='49' OR pp.Size='50' OR pp.Size='51' OR pp.Size='52' THEN '48-52 CM'
                  WHEN pp.Size='54' OR pp.Size='55' OR pp.Size='56' OR pp.Size='57' OR pp.Size='58' THEN '54-58 CM'
                  WHEN pp.Size='60' OR pp.Size='61' OR pp.Size='62' THEN '60-62 CM'
                  WHEN pp.Size='70' THEN '70 CM'
                  WHEN pp.Size='S' THEN 'S'
                  WHEN pp.Size='M' THEN 'M'
                  WHEN pp.Size='L' THEN 'L'
                  WHEN pp.Size='XL' THEN 'XL'
                  ELSE 'NA'
                  END
                  AS SizeRange,
                  pp.ListPrice, pp.StandardCost, pp.ReorderPoint,
                  pp.SafetyStockLevel, pp.Color, pp.FinishedGoodsFlag, pp.MakeFlag, pp.ProductNumber, pp.Name AS ProductName, pp.ProductID

FROM          Production.Product pp LEFT JOIN
                    Production.ProductModel ppm ON pp.ProductModelID = ppm.ProductModelID;
```

1	**Table Name**	**DimProduct**														
2	**Table Type**	**Dimension**														
3	**Display Name**	**Product Dimension**														
4			**Target**									**Source**				**ETL**
5	**Database**	**Schema**	**Table Name**	**Column Name**	**Data Type**	**Size**	**Key?**	**FK To**	**NULL?**	**SCD Type**	**Database**	**Schema**	**Table/View Name**	**Column Name**	**Data Type**	**Size**	**ETL Rules**
6	AdventureWorksDW	dbo	DimProduct	ProductKey	int		PK ID		NO		Derived						Generated
7	AdventureWorksDW	dbo	DimProduct	BusinessKey	int				NO	BusinessKey	AdventureWorks2019	Production	Product	ProductID	int		Get from source
8	AdventureWorksDW	dbo	DimProduct	ProductAlternateKey	nvarchar	25			YES		AdventureWorks2019	Production	Product	ProductNumber	nvarchar	25	Get from source
9	AdventureWorksDW	dbo	DimProduct	ProductSubcategoryKey	int		FK	DimProc	YES		AdventureWorksDW	dbo	DimProductSubcategory	ProductSubcategoryKey	int		Get from DimProductSubcategory
10	AdventureWorksDW	dbo	DimProduct	ProductName	nvarchar	50			NO		AdventureWorks2019	Production	Product	Name	nvarchar	50	Direct move
11	AdventureWorksDW	dbo	DimProduct	Color	nvarchar	15			YES	2	AdventureWorks2019	Production	Product	Color	nvarchar	15	Direct move
12	AdventureWorksDW	dbo	DimProduct	ListPrice	money				YES	2	AdventureWorks2019	Production	Product	ListPrice	money		Direct move
13	AdventureWorksDW	dbo	DimProduct	SizeRange	nvarchar	50			YES		AdventureWorks2019	Production	Product	Size	nvarchar	5	Derived from column Size
14	AdventureWorksDW	dbo	DimProduct	ProductLine	nchar	2			YES		AdventureWorks2019	Production	Product	ProductLine	nchar	2	Direct move
15	AdventureWorksDW	dbo	DimProduct	Class	nchar	2			YES		AdventureWorks2019	Production	Product	Class	nchar	2	Direct move
16	AdventureWorksDW	dbo	DimProduct	Style	nchar	2			YES		AdventureWorks2019	Production	Product	Style	nchar	2	Direct move
17	AdventureWorksDW	dbo	DimProduct	ModelName	nvarchar	50			YES		AdventureWorks2019	Production	ProductModel	Name	nvarchar	50	Direct move
18	AdventureWorksDW	dbo	DimProduct	StartDate	datetime				YES		Derived						SCD-2
19	AdventureWorksDW	dbo	DimProduct	EndDate	datetime				YES		Derived						SCD-2
20	AdventureWorksDW	dbo	DimProduct	Status	nvarchar	7			YES		Derived						SCD-2

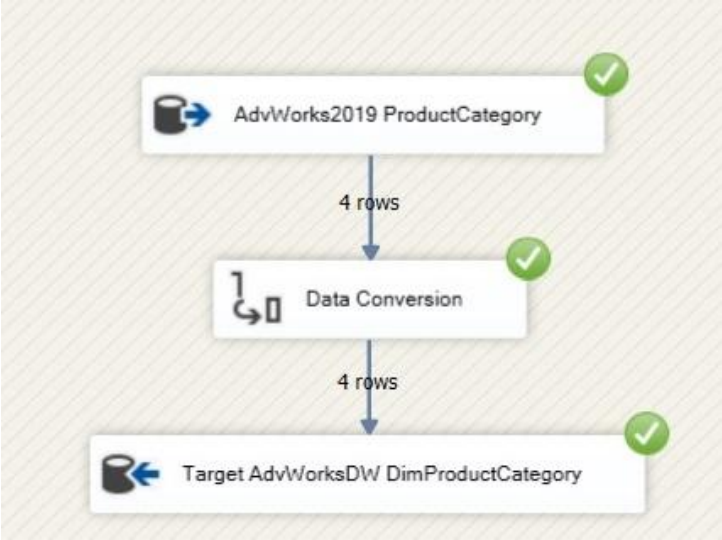
Source to Target Mapping
- For FactInternetSales

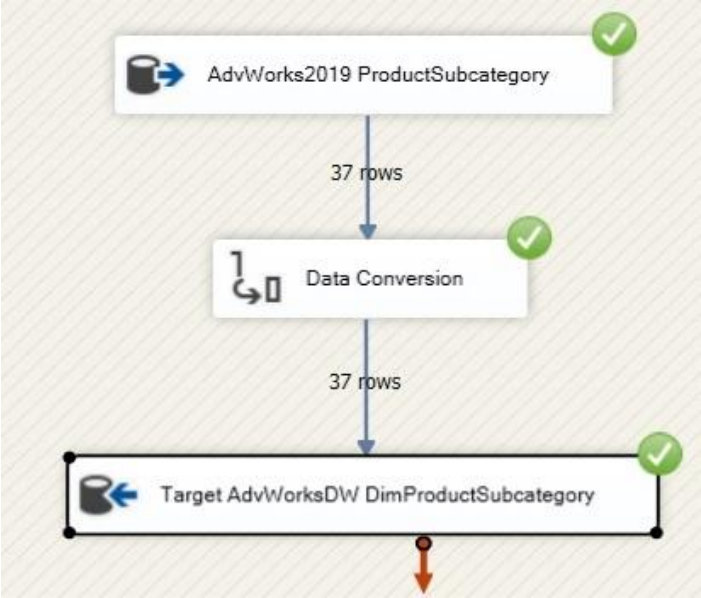
```sql
USE AdventureWorks2019
GO

CREATE View dbo.etl_FactInternetSales AS
SELECT soh.[SalesOrderNumber],
    CAST( ROW_NUMBER() OVER (Partition by soh.[SalesOrderNumber] ORDER BY sod.SalesOrderDetailID) AS tinyint) AS LineNumber,
    CAST( [OrderDate] AS Date) AS OrderDate,
    CAST( [ShipDate]  AS Date) AS ShipDate,
    p.ProductNumber AS ProductID,
    CAST( c.AccountNumber as nvarchar(10)) AS AccountNumber, soh.[TerritoryID],
-- measures
    sod.OrderQty, sod.LineTotal AS [SubTotal], sod.UnitPrice AS UnitPrice      ,[TaxAmt]      ,[Freight]      ,[TotalDue],
    p.StandardCost,
-- order fact columns
    soh.[PurchaseOrderNumber],
        psp.CountryRegionCode,
        CAST (CASE psp.CountryRegionCode
        WHEN 'US' THEN 'USD'
        WHEN 'CA' THEN 'CAD'
        WHEN 'FR' THEN 'FRF'
        WHEN 'DE' THEN 'DEM'
        WHEN 'GB' THEN 'GBP'
        ELSE 'AUD'
        END
        as nchar(3)) AS CurrencyCode,
        sod.SpecialOfferID
  FROM Sales.SalesOrderDetail sod
    INNER JOIN [Sales].[SalesOrderHeader] soh
      ON soh.SalesOrderID = sod.SalesOrderID
    INNER JOIN Sales.Customer c
      ON c.CustomerID = soh.CustomerID
    INNER JOIN Production.Product p
      ON p.ProductID = sod.ProductID
        INNER JOIN Person.Address pa
    ON soh.BillToAddressID=pa.AddressID
    INNER JOIN Person.StateProvince psp
    ON pa.StateProvinceID=psp.StateProvinceID
    where OnlineOrderFlag = 1 ;
```

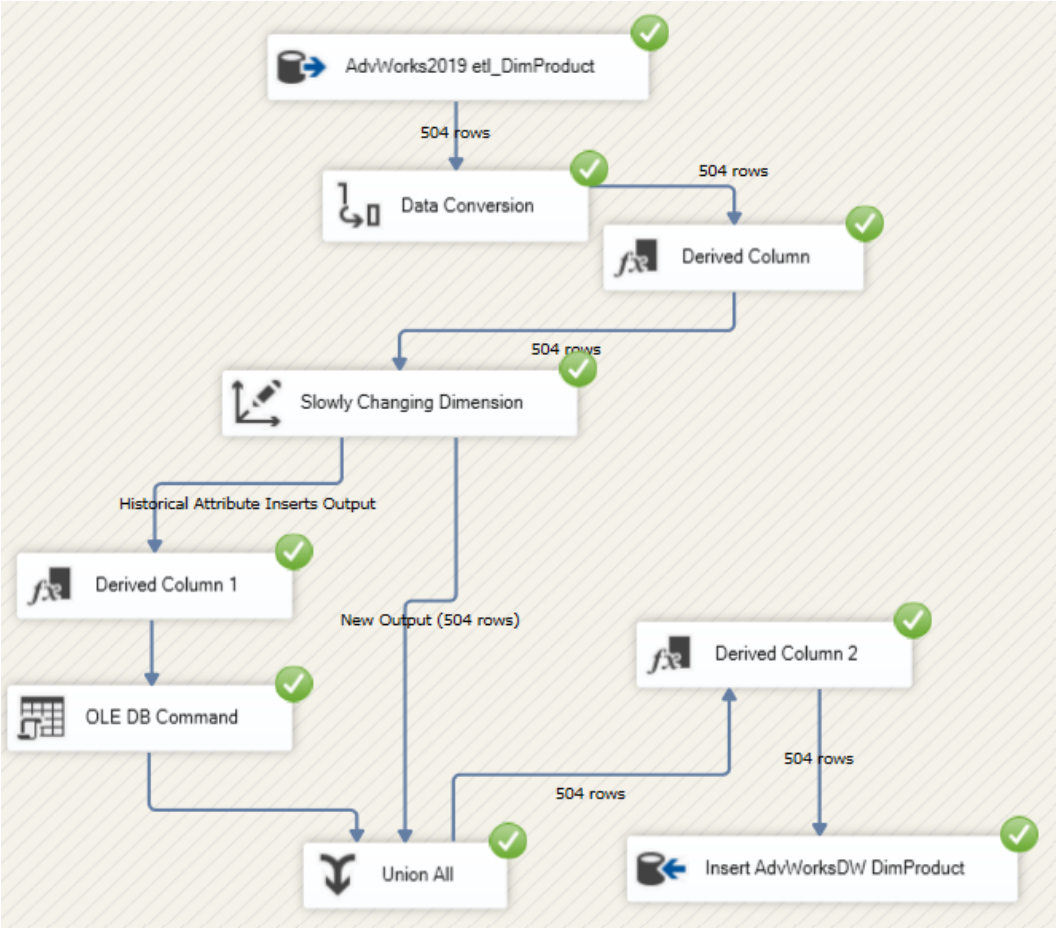
1	**Table Name**	**FactInternetSales**													
2	**Table Type**	**Fact**													
3	**Display Name**	**FactInternetSales**													
4				**Target**							**Source**				**ETL**
5	**Database**	**Schema**	**Table Name**	**Column Name**	**Data Type**	**Size**	**Key?**	**FK To**	**Nullable**	**Database**	**Schema**	**Table/View Name**	**Column Name**	**Data Type**	**Size**	**ETL Rules**
6	AdventureWorksDW	dbo	FactInternetSales	ProductKey	int		FK	DimProduc	NO	AdventureWorksDW	dbo	DimProduct	ProductKey	int		key lookup from DimProduct
7	AdventureWorksDW	dbo	FactInternetSales	PromotionKey	int		FK	DimPromo	NO	AdventureWorksDW	dbo	DimPromotion	PromotionKey	int		key lookup from DimPromotion
8	AdventureWorksDW	dbo	FactInternetSales	OrderDateKey	int		FK	DimGeogr	NO	AdventureWorksDW	dbo	DimDate	DateKey	int		key lookup from DimDate
9	AdventureWorksDW	dbo	FactInternetSales	ShipDateKey	int		FK	DimDate	NO	AdventureWorksDW	dbo	DimDate	DateKey	int		key lookup from DimDate
10	AdventureWorksDW	dbo	FactInternetSales	CustomerKey	int		FK	DimCustor	NO	AdventureWorksDW	dbo	DimCustomer	CustomerKey	int		key lookup from DimCustomer
11	AdventureWorksDW	dbo	FactInternetSales	SalesTerritoryKey	int		FK	DimSalesT	NO	AdventureWorksDW	dbo	DimSalesTerritory	SalesTerritoryKey	int		key lookup from DimSalesTerrritory
12	AdventureWorksDW	dbo	FactInternetSales	CurrencyKey	int		FK	DimCurren	NO	AdventureWorksDW	dbo	DimCurrency	CurrencyKey	int		key lookup from DimCurrency
13	AdventureWorksDW	dbo	FactInternetSales	SalesOrderNumber	nvarchar	20	PK		NO	AdventureWorks2019	Sales	SalesOrderHeader	SalesOrderNumber	nvarchar	25	Direct move
14	AdventureWorksDW	dbo	FactInternetSales	SalesOrderLineNumber	tinyint		PK		NO	AdventureWorks2019	Sales	SalesOrderDetail	SaleOrderDetailID	int		Direct move
15	AdventureWorksDW	dbo	FactInternetSales	OrderQuantity	smallint				NO	AdventureWorks2019	Sales	SalesOrderDetail	OrderQty	smallint		Direct move
16	AdventureWorksDW	dbo	FactInternetSales	UnitPrice	money				NO	AdventureWorks2019	Sales	SalesOrderDetail	UnitPrice	money		Direct move
17	AdventureWorksDW	dbo	FactInternetSales	ExtendedAmount	money				NO	AdventureWorks2019	Sales	SalesOrderDetail	LIneTotal	numeric	38,6	Data conversion
18	AdventureWorksDW	dbo	FactInternetSales	TotalProductCost	money				NO	AdventureWorks2019	Production	Product	StandardCost	money		Direct move
19	AdventureWorksDW	dbo	FactInternetSales	SalesAmount	money				NO	AdventureWorks2019	Sales	SalesOrderDetail	LIneTotal	numeric	38,6	Data conversion
20	AdventureWorksDW	dbo	FactInternetSales	CarrierTrackingNumber	nvarchar	25			YES	AdventureWorks2019	Sales	SalesOrderDetail	CarrierTrackingNumber	nvarchar	25	Direct move
21	AdventureWorksDW	dbo	FactInternetSales	CustomerPONumber	nvarchar	25			YES	AdventureWorks2019	Sales	SalesOrderHeader	PurchaseOrderNumber	nvarchar	25	Direct move

Load DimProduct – Load Referenced Tables DimProductCategory and DimProductSubcategroy





Load DimProduct – SCD Type2
(History will be added as a new row)



Load DimProduct Cont.

- ETL Result: 504 rows in DimProduct

```
SELECT * FROM [dbo].[DimProduct];
```

100 %

Results | Messages

	ProductKey	BusinessKey	ProductAlternateKey	ProductSubcategoryKey	ProductName	Color	ListPrice	Size	SizeRange	ProductLine	Class	Style	ModelName	StartDate	EndDate	Status
1	1	1	AR-5381	NULL	Adjustable Race	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
2	2	2	BA-8327	NULL	Bearing Ball	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
3	3	3	BE-2349	NULL	BB Ball Bearing	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
4	4	4	BE-2908	NULL	Headset Ball Bearings	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
5	5	316	BL-2036	NULL	Blade	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
6	6	317	CA-5965	NULL	LL Crankarm	Black	0.00	NULL	NA	NULL	L	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
7	7	318	CA-6738	NULL	ML Crankarm	Black	0.00	NULL	NA	NULL	M	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
8	8	319	CA-7457	NULL	HL Crankarm	Black	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
9	9	320	CB-2903	NULL	Chainring Bolts	Silver	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
10	10	321	CN-6137	NULL	Chainring Nut	Silver	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
11	11	322	CR-7833	NULL	Chainring	Black	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
12	12	323	CR-9981	NULL	Crown Race	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
13	13	324	CS-2812	NULL	Chain Stays	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
14	14	325	DC-8732	NULL	Decal 1	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
15	15	326	DC-9824	NULL	Decal 2	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
16	16	327	DT-2377	NULL	Down Tube	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
17	17	328	EC-M092	NULL	Mountain End Caps	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
18	18	329	EC-R098	NULL	Road End Caps	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
19	19	330	EC-T209	NULL	Touring End Caps	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
20	20	331	FE-3760	NULL	Fork End	NULL	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL
21	21	332	FH-2981	NULL	Freewheel	Silver	0.00	NULL	NA	NULL	NULL	NULL	NULL	2022-10-18 17:14:02.000	NULL	NULL

Query executed successfully. HUOGANG2022 (15.0 RTM) sa (52) AdventureWorksDW101 00:00:00 504 rows

SCD Validation For DimProduct

Before

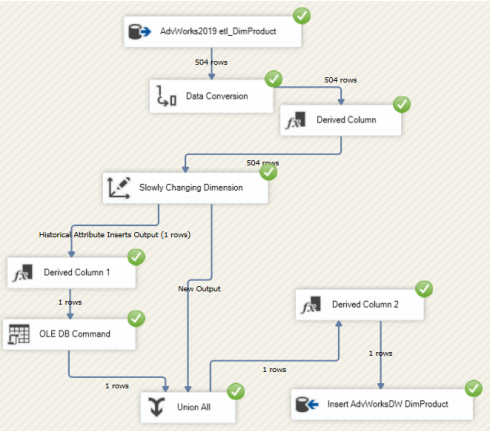
```
SELECT * FROM [dbo].[DimProduct] WHERE [BusinessKey]=800
```

100 %

Results | Messages

	ProductKey	BusinessKey	ProductAlternateKey	ProductSubcategoryKey	ProductName	Color	ListPrice	Size	SizeRange	ProductLine	Class	Style	ModelName	StartDate	EndDate	Status
1	305	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Yellow	1120.49	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:14:02.000	NULL	NULL

```
UPDATE [Production].[Product] SET Color='Red' WHERE [ProductID]=800;
```



After

```
SELECT * FROM [dbo].[DimProduct] WHERE [BusinessKey]=800;
```

100 %

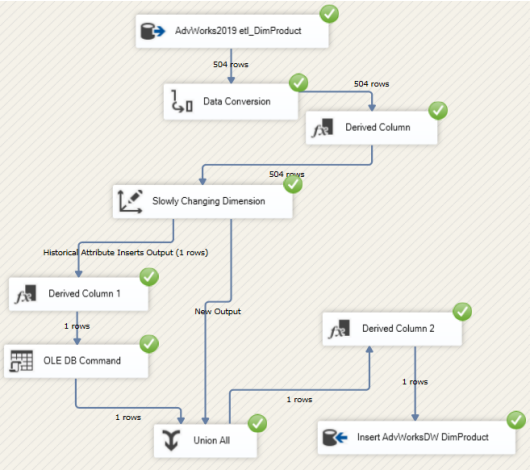
Results | Messages

	ProductKey	BusinessKey	ProductAlternateKey	ProductSubcategoryKey	ProductName	Color	ListPrice	Size	SizeRange	ProductLine	Class	Style	ModelName	StartDate	EndDate
1	305	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Yellow	1120.49	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:14:02.000	2022-10-18 17:25:30.00
2	505	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Red	1120.49	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:25:30.000	NULL

SCD Validation For DimProduct Cont.

Before

```
UPDATE [Production].[Product] SET ListPrice=1200 WHERE [ProductID]=800;
```



After

```
SELECT * FROM [dbo].[DimProduct] WHERE [BusinessKey]=800;
```

100 %

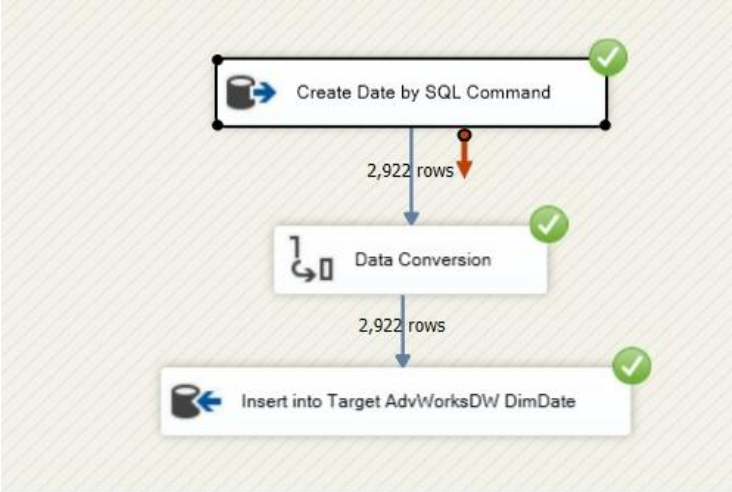
Results | Messages

	ProductKey	BusinessKey	ProductAlternateKey	ProductSubcategoryKey	ProductName	Color	ListPrice	Size	SizeRange	ProductLine	Class	Style	ModelName	StartDate	EndDate
1	305	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Yellow	1120.49	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:14:02.000	2022-10-18 17:25:30.0
2	505	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Red	1120.49	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:25:30.000	2022-10-18 17:33:07.0
3	506	800	BK-R64Y-44	2	Road-550-W Yellow, 44	Red	1200.00	44	42-46 CM	R	M	W	Road-550-W	2022-10-18 17:33:07.000	NULL

Load DimDate – Derived Table

```sql
DECLARE @startdate DATE = '20100101'
    , @enddate DATE = '20171231' ;

;WITH c
AS (
    SELECT  Num = ROW_NUMBER() OVER (ORDER BY (SELECT NULL)) - 1
    FROM sys.columns c
    CROSS JOIN sys.columns c1
    )
, d
AS (
    SELECT  [date] = DATEADD(day, Num, @startdate), Num
    FROM c
    WHERE Num >= 0 AND Num <= DATEDIFF(day, @startdate, @enddate)
    )
SELECT DateKey = CAST(CONVERT(VARCHAR(8), DATEADD(day, Num, @startdate), 112) AS INT)
    , [date] AS FullDateAlternateKey
    , [DayNumberOfWeek] = DATEPART(weekday, [Date])
    , [DayNameOfWeek] = DATENAME(weekday, [Date])
    , [DayNumberOfMonth] = DATEPART(day, [Date])
    , [DayNumberOfYear] = DATEPART(dayofyear, [Date])
    , [WeekNumberOfYear] = DATEPART(week, [Date])
    , [MonthName] = DATENAME(month, [Date])
    , [MonthNumberOfYear] = DATEPART(month, [Date])
    , [CalendarQuarter] = DATEPART(quarter, [Date])
    , [CalendarYear] = YEAR([date])
    , [CalendarSemester] = ((DATEPART(quarter,[Date])-1)/2)+1
--Add more columns as needed.
FROM d
```




```sql
USE AdventureWorksDW;
SELECT * FROM DimDate;
```

2010-01-01 to 2017-12-31: 2922 rows

100 %

Results | Messages

	DateKey	FullDateAlternateKey	DayNumberOfWeek	DayNameOfWeek	DayNumberOfMonth	DayNumberOfYear	WeekNumberOfYear	MonthName	MonthNumberOfYea
1	20100101	2010-01-01	6	Friday	1	1	1	January	1
2	20100102	2010-01-02	7	Saturday	2	2	1	January	1
3	20100103	2010-01-03	1	Sunday	3	3	2	January	1
4	20100104	2010-01-04	2	Monday	4	4	2	January	1
5	20100105	2010-01-05	3	Tuesday	5	5	2	January	1
6	20100106	2010-01-06	4	Wednesday	6	6	2	January	1
7	20100107	2010-01-07	5	Thursday	7	7	2	January	1
8	20100108	2010-01-08	6	Friday	8	8	2	January	1
9	20100109	2010-01-09	7	Saturday	9	9	2	January	1
10	20100110	2010-01-10	1	Sunday	10	10	3	January	1
11	20100111	2010-01-11	2	Monday	11	11	3	January	1
12	20100112	2010-01-12	3	Tuesday	12	12	3	January	1

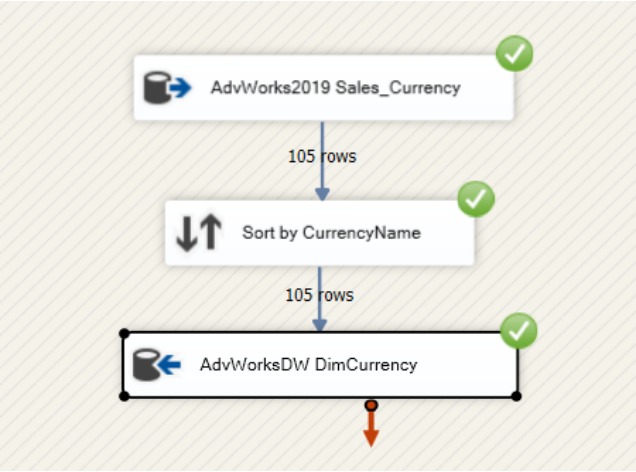
Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (66) | AdventureWorksDW | 00:00:00 | 2,922 rows

Validation of DimDate By Excel

- 2010-01-01 to 2017-12-31: 2922 rows

	A	B	C	D	E	F	G	H	I	J	K	L
1	DateKey	FullDateAlternateKey	DayNumberOfWeek	DayNameOf Week	DayNumberOfMonth	DayNumberOfYear	WeekNumberOfYear	MonthName	MonthNumberOfYear	CalendarQuarter	CalendarYear	CalendarSemester
2	20100101	2010-01-01	6	Friday	1	1	1	January	1	1	2010	1
3	20100102	2010-01-02	7	Saturday	2	2	1	January	1	1	2010	1
4	20100103	2010-01-03	1	Sunday	3	3	2	January	1	1	2010	1
5	20100104	2010-01-04	2	Monday	4	4	2	January	1	1	2010	1
6	20100105	2010-01-05	3	Tuesday	5	5	2	January	1	1	2010	1
7	20100106	2010-01-06	4	Wednesday	6	6	2	January	1	1	2010	1
8	20100107	2010-01-07	5	Thursday	7	7	2	January	1	1	2010	1
9	20100108	2010-01-08	6	Friday	8	8	2	January	1	1	2010	1
10	20100109	2010-01-09	7	Saturday	9	9	2	January	1	1	2010	1
11	20100110	2010-01-10	1	Sunday	10	10	3	January	1	1	2010	1
12	20100111	2010-01-11	2	Monday	11	11	3	January	1	1	2010	1
13	20100112	2010-01-12	3	Tuesday	12	12	3	January	1	1	2010	1
14	20100113	2010-01-13	4	Wednesday	13	13	3	January	1	1	2010	1
2912	20171220	2017-12-20	4	Wednesday	20	354	51	December	12	4	2017	2
2913	20171221	2017-12-21	5	Thursday	21	355	51	December	12	4	2017	2
2914	20171222	2017-12-22	6	Friday	22	356	51	December	12	4	2017	2
2915	20171223	2017-12-23	7	Saturday	23	357	51	December	12	4	2017	2
2916	20171224	2017-12-24	1	Sunday	24	358	52	December	12	4	2017	2
2917	20171225	2017-12-25	2	Monday	25	359	52	December	12	4	2017	2
2918	20171226	2017-12-26	3	Tuesday	26	360	52	December	12	4	2017	2
2919	20171227	2017-12-27	4	Wednesday	27	361	52	December	12	4	2017	2
2920	20171228	2017-12-28	5	Thursday	28	362	52	December	12	4	2017	2
2921	20171229	2017-12-29	6	Friday	29	363	52	December	12	4	2017	2
2922	20171230	2017-12-30	7	Saturday	30	364	52	December	12	4	2017	2
2923	20171231	2017-12-31	1	Sunday	31	365	53	December	12	4	2017	2

Load DimCurrency



ETL Result: 105 rows in DimCurrency

```
USE AdventureWorksDW
GO
SELECT * FROM [dbo].[DimCurrency];
```

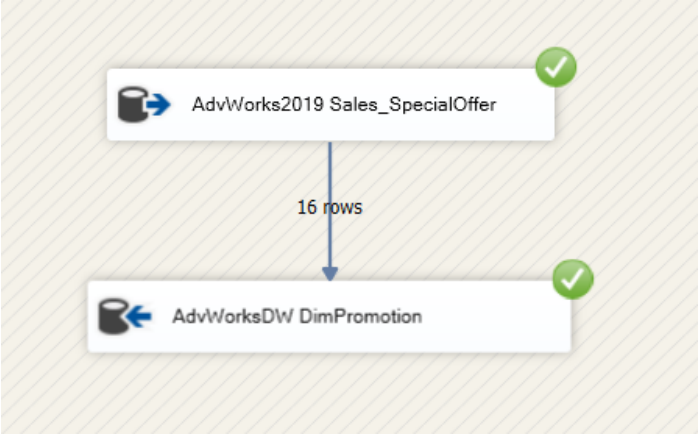
100 %

Results | Messages

	CurrencyKey	CurrencyAlternateKey	CurrencyName
1	1	AFA	Afghani
2	2	DZD	Algerian Dinar
3	3	ARS	Argentine Peso
4	4	AMD	Armenian Dram
5	5	AWG	Aruban Guilder
6	6	AUD	Australian Dollar
7	7	AZM	Azerbaijanian Manat
8	8	BSD	Bahamian Dollar
9	9	BHD	Bahraini Dinar
10	10	THB	Baht
11	11	PAB	Balboa
12	12	BBD	Barbados Dollar

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (53) | AdventureWorksDW | 00:00:00 | 105 rows

Load DimPromotion



ETL Result: 16 rows in DimPromotion

```
USE AdventureWorksDW
GO
SELECT * FROM [dbo].[DimPromotion];
```

100 %

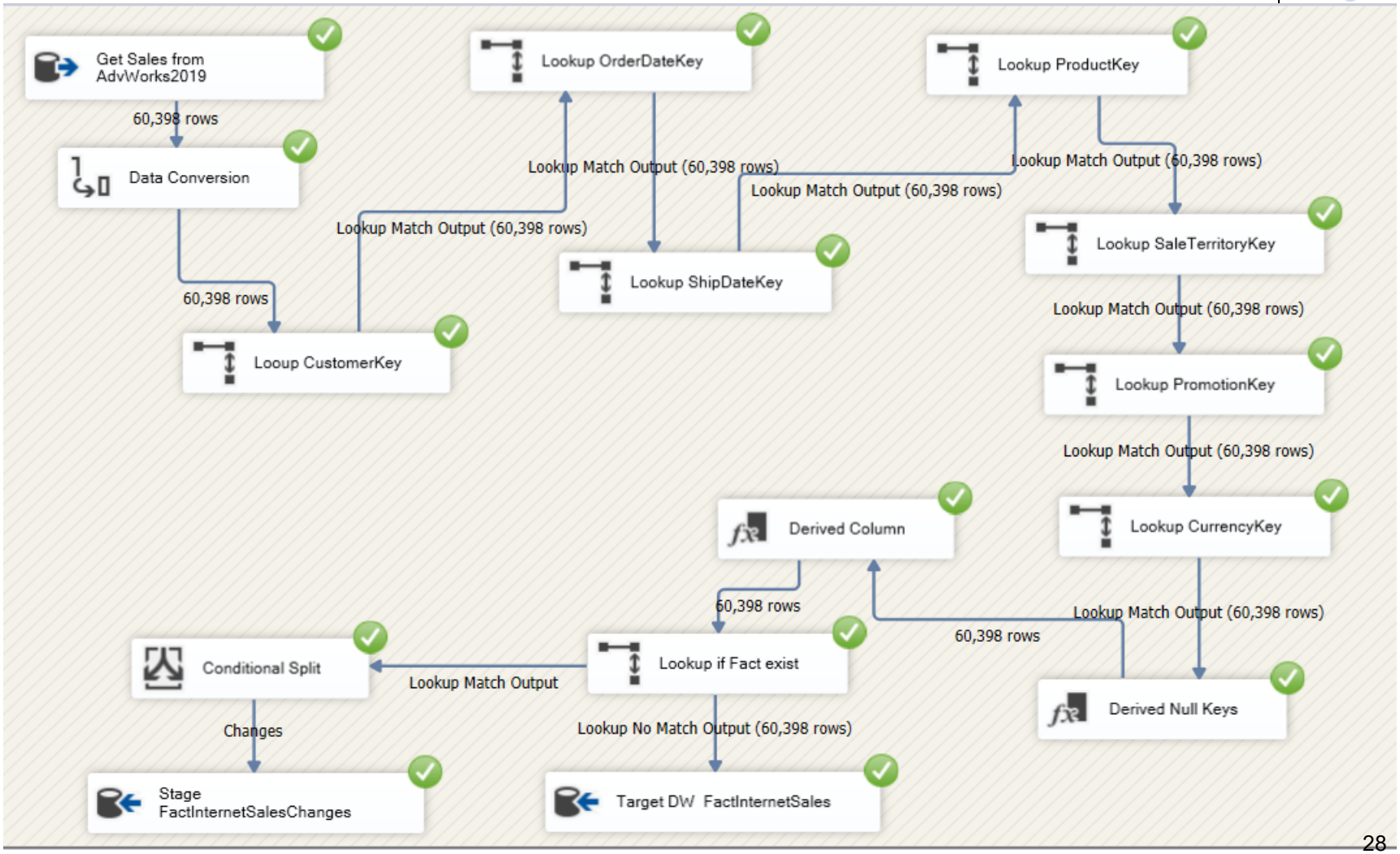
Results | Messages

	PromotionKey	PromotionID	PromotionName	DiscountPct	PromotionType	PromotionCategory	StartDate	EndDate
1	1	1	No Discount	0	No Discount	No Discount	2011-05-01 00:00:00.000	2014-11-30 00:00
2	2	2	Volume Discount 11 to 14	0.02	Volume Discount	Reseller	2011-05-31 00:00:00.000	2014-05-30 00:00
3	3	3	Volume Discount 15 to 24	0.05	Volume Discount	Reseller	2011-05-31 00:00:00.000	2014-05-30 00:00
4	4	4	Volume Discount 25 to 40	0.1	Volume Discount	Reseller	2011-05-31 00:00:00.000	2014-05-30 00:00
5	5	5	Volume Discount 41 to 60	0.15	Volume Discount	Reseller	2011-05-31 00:00:00.000	2014-05-30 00:00
6	6	6	Volume Discount over 60	0.2	Volume Discount	Reseller	2011-05-31 00:00:00.000	2014-05-30 00:00
7	7	7	Mountain-100 Clearance Sale	0.35	Discontinued Product	Reseller	2012-04-13 00:00:00.000	2012-05-29 00:00
8	8	8	Sport Helmet Discount-2002	0.1	Seasonal Discount	Reseller	2012-05-30 00:00:00.000	2012-06-29 00:00
9	9	9	Road-650 Overstock	0.3	Excess Inventory	Reseller	2012-05-30 00:00:00.000	2012-07-30 00:00
10	10	10	Mountain Tire Sale	0.5	Excess Inventory	Customer	2013-05-14 00:00:00.000	2013-07-29 00:00
11	11	11	Sport Helmet Discount-2003	0.15	Seasonal Discount	Reseller	2013-05-30 00:00:00.000	2013-06-29 00:00
12	12	12	LL Road Frame Sale	0.35	Excess Inventory	Reseller	2013-05-30 00:00:00.000	2013-07-14 00:00

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (53) | AdventureWorksDW | 00:00:00 | 16 rows

Load FactInternetSales

- SSIS Package First Execution



Load FactInternetSales Cont.
- SSIS Package Execution Result

- ETL Result: 60,398 rows

SQLQuery5.sql - H...eWorksDW (sa (54))* | SQLQuery4.sql - H...eWorksDW (sa (53)) | SQLQuery3.sql - H...eWorksDW (sa (65)) | SQLQuery2.sql - H...eWorksDW (sa (55))*

```
USE AdventureWorksDW

SELECT * FROM [dbo].[FactInternetSales]
```

100 %

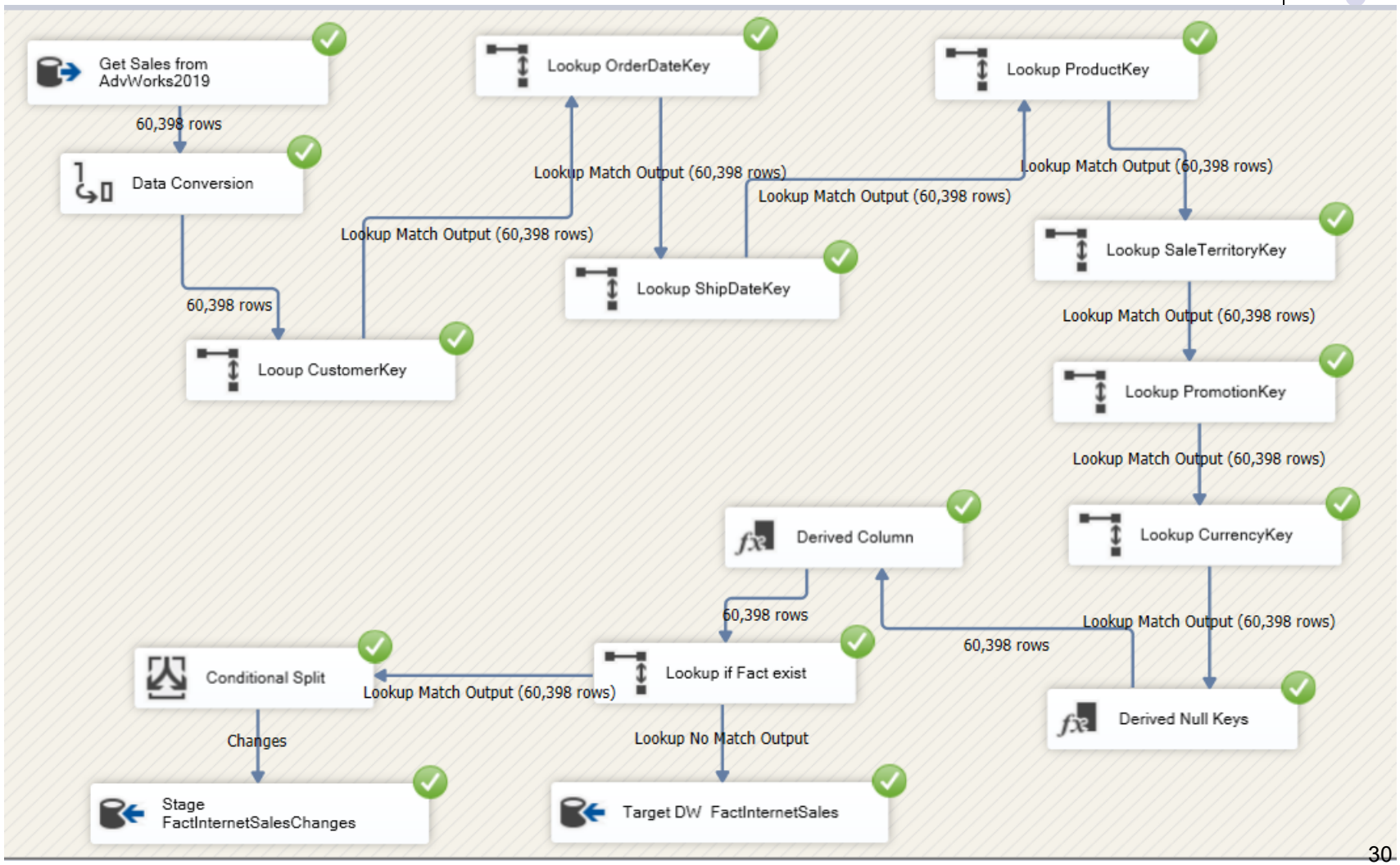
Results | Messages

	ProductKey	PromotionKey	OrderDateKey	ShipDateKey	CustomerKey	CurrencyKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	OrderQuantity	UnitPrice	ExtendedAmount	TotalProductCost	SalesAm
1	254	1	20110531	20110607	21768	19	6	SO43697	1	1	3578.27	3578.27	2171.2942	3578.27
2	278	1	20110531	20110607	28389	39	7	SO43698	1	1	3399.99	3399.99	1912.1544	3399.99
3	278	1	20110531	20110607	25863	100	1	SO43699	1	1	3399.99	3399.99	1912.1544	3399.99
4	272	1	20110531	20110607	14501	100	4	SO43700	1	1	699.0982	699.0982	486.7066	699.098
5	278	1	20110531	20110607	11003	6	9	SO43701	1	1	3399.99	3399.99	1912.1544	3399.99
6	255	1	20110601	20110608	27645	100	4	SO43702	1	1	3578.27	3578.27	2171.2942	3578.27
7	254	1	20110601	20110608	16624	6	9	SO43703	1	1	3578.27	3578.27	2171.2942	3578.27
8	283	1	20110601	20110608	11005	6	9	SO43704	1	1	3374.99	3374.99	1898.0944	3374.99
9	276	1	20110601	20110608	11011	6	9	SO43705	1	1	3399.99	3399.99	1912.1544	3399.99
10	256	1	20110602	20110609	27621	100	4	SO43706	1	1	3578.27	3578.27	2171.2942	3578.27
11	256	1	20110602	20110609	27616	100	4	SO43707	1	1	3578.27	3578.27	2171.2942	3578.27
12	269	1	20110602	20110609	20042	98	10	SO43708	1	1	699.0982	699.0982	486.7066	699.098
13	257	1	20110602	20110609	16351	6	9	SO43709	1	1	3578.27	3578.27	2171.2942	3578.27
14	258	1	20110602	20110609	16517	6	9	SO43710	1	1	3578.27	3578.27	2171.2942	3578.27
15	258	1	20110603	20110610	27606	100	1	SO43711	1	1	3578.27	3578.27	2171.2942	3578.27
16	255	1	20110603	20110610	13513	29	8	SO43712	1	1	3578.27	3578.27	2171.2942	3578.27
17	254	1	20110604	20110611	27601	100	4	SO43713	1	1	3578.27	3578.27	2171.2942	3578.27
18	255	1	20110604	20110611	13591	98	10	SO43714	1	1	3578.27	3578.27	2171.2942	3578.27
19	258	1	20110604	20110611	16483	6	9	SO43715	1	1	3578.27	3578.27	2171.2942	3578.27
20	255	1	20110604	20110611	16529	6	9	SO43716	1	1	3578.27	3578.27	2171.2942	3578.27
21	272	1	20110604	20110611	25249	6	9	SO43717	1	1	699.0982	699.0982	486.7066	699.098

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (54) | AdventureWorksDW | 00:00:00 | 60,398 rows

Load FactInternetSales

- SSIS Package Second Execution



Load FactInternetSales Cont.

- Data Validation

- ❑ **Any Missing Data?**

AdventureWorksDW vs AdventureWorksDW2019

```sql
USE AdventureWorksDW
GO
SELECT 'FactInternetSales Total Rows Counts', COUNT(*)
FROM [dbo].[FactInternetSales]
UNION
SELECT 'DISTINCT CustomerKey Counts',COUNT(DISTINCT [CustomerKey] )
FROM [dbo].[FactInternetSales]
UNION
SELECT 'DISTINCT SalesOrderNumber Counts',COUNT(DISTINCT [SalesOrderNumber])
FROM [dbo].[FactInternetSales];

USE AdventureWorksDW2019
GO
SELECT 'FactInternetSales Total Rows Counts', COUNT(*)
FROM [dbo].[FactInternetSales]
UNION
SELECT 'DISTINCT CustomerKey Counts',COUNT(DISTINCT [CustomerKey] )
FROM [dbo].[FactInternetSales]
UNION
SELECT 'DISTINCT SalesOrderNumber Counts',COUNT(DISTINCT [SalesOrderNumber])
FROM [dbo].[FactInternetSales]
```

90 %

Results | Messages

	(No column name)	(No column name)
1	FactInternetSales Total Rows Counts	60398
2	DISTINCT CustomerKey Counts	18484
3	DISTINCT SalesOrderNumber Counts	27659

	(No column name)	(No column name)
1	FactInternetSales Total Rows Counts	60398
2	DISTINCT CustomerKey Counts	18484
3	DISTINCT SalesOrderNumber Counts	27659

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (60) | AdventureWorksDW2019 | 00:00:00 | 6 rows

Load FactInternetSales Cont.

- Data Validation

- Any duplicated records?

```sql
USE AdventureWorksDW
GO

SELECT [SalesOrderNumber],[SalesOrderLineNumber],COUNT(*)
FROM [dbo].[FactInternetSales]
GROUP BY [SalesOrderNumber],[SalesOrderLineNumber]
HAVING COUNT(*)>1
```

100 %

Results | Messages

SalesOrderNumber	SalesOrderLineNumber	(No column name)

Query executed successfully. | HUOGANG2022 (15.0 RTM) | sa (60) | AdventureWorksDW | 00:00:00 | 0 rows